# DICTIONARY

- If we want to represent a group of objects as key-value pairs then we should go for Dictionary.
- Duplicate keys are not allowed but values can be duplicated.
- Heterogeneous objects are allowed for both key and values.
- Insertion order is not preserved
- Dictionaries are mutable
- Dictionaries are dynamic
- indexing and slicing concepts are not applicable

syntax => d = {key:value, key:value}

## How to Create Dictionary?

- d = {} OR d = dict()

- If we know data in advance, then we can create dictionary as follows

  d = {100:’python’ , 200:'java', 300:'Typescript'}

## How to Access Data from the Dictionary?

We can access data by using keys.

d = {100:’python’ , 200:'java', 300:'Typescript'}

print(d[100]) → python

print(d[400]) → KeyError

## How to Update Dictionaries?

- d[key] = value
- If the key is not available, then a new entry will be added to the dictionary with the specified key-value pair
- If the key is already available, then old value will be replaced with new value.

```
d={100:"python",200:"java",300:"javascript"}
print(d)
d[400]="typescript"
print(d)
d[100]="fortran"
print(d)
```

## How to Delete Elements from Dictionary?

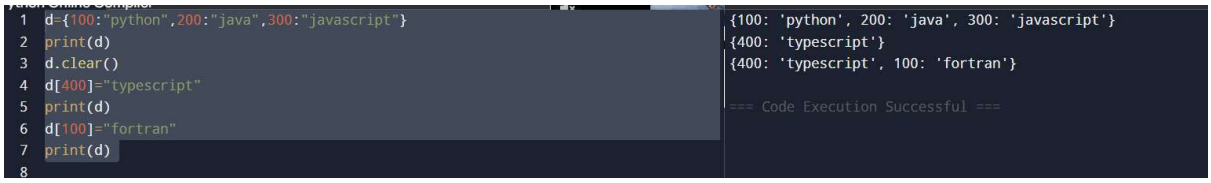
1. **del d[key]**
   - It deletes entry associated with the specified key.
   - If the key is not available, then we will get KeyError.

```
d={100:"python",200:"java",300:"javascript"}
print(d)
del d[100]
d[400]="typescript"
print(d)
d[100]="fortran"
print(d)
```

## 2. d.clear()

To remove all entries from the dictionary.

```
d={100:"python",200:"java",300:"javascript"}
print(d)
d.clear()
d[400]="typescript"
print(d)
d[100]="fortran"
print(d)
```



## 3. **del d**

delete total dictionary.

Now we cannot access d.

```
d={100:"python",200:"java",300:"javascript"}
print(d)
del d
print(d)
```

## Important Functions of Dictionary:

1) **dict():**
   **To create a dictionary**

2) **len()**
   **Returns the number of items in the dictionary.**

```
inventory = {"apples": 5, "bananas": 12, "oranges": 8}
count = len(inventory)
print(count) →# Output: 3
```

3) **clear():**
   **To remove all elements from the dictionary.**

```
car = {  "brand": "Ford",   "model": "Mustang",  "year": 1964}
car.clear()
print(car) → # Output: {}
```

4) **get():**
   **To get the value associated with the key**

a. ***d.get(key) → If the key is available then returns the corresponding value otherwise returns None. It won't raise any error.***

b. ***d.get(key, defaultvalue) → If the key is available then returns the corresponding value otherwise returns default value.***

```
# Initial dictionary
student = {"name": "Alice", "grade": "A"}

print(student.get("name"))    # Output: Alice
print(student.get("age"))     # Output: None (no error raised)
```

```
print(student.get("grade", "N/A")) # Output: A
print(student.get("age", 0))      # Output: 0 (uses default because key is missing)
```

## 5) pop():

### d.pop(key)

- **It removes the entry associated with the specified key and returns the corresponding value.**
- **If the specified key is not available, then we will get KeyError.**

```
user_data = {"id": 101, "username": "coder_99", "status": "active"}
status_value = user_data.pop("status")
print(status_value)  # Output: active
print(user_data)     # Output: {'id': 101, 'username': 'coder_99'}

# Key not available: raises a KeyError
# user_data.pop("email")  # This would raise: KeyError: 'email'
```

## 6) popitem():

### It removes an arbitrary item(key-value) from the dictionary and returns it.

```
profile = {"username": "jdoe", "email": "jd@example.com", "level": 5}
item = profile.popitem()
print(item)    # Output: ('level', 5)
print(profile) # Output: {'username': 'jdoe', 'email': 'jd@example.com'}
```

## 7) keys():

**It returns all keys associated with dictionary.**

```
device = {"brand": "Apple", "model": "iPad", "storage": "256GB"}
all_keys = device.keys()
print(all_keys) → Output: dict_keys(['brand', 'model', 'storage'])
# Convert to a list if needed
keys_list = list(device.keys())
print(keys_list) → # Output: ['brand', 'model', 'storage']
```

## 8) values():

**It returns all values associated with the dictionary.**

```
device = {"brand": "Apple", "model": "iPad", "storage": "256GB"}
all_values = device.values()
print(all_values)
# Convert to a list if needed
keys_list = list(device.values())
print(keys_list)
# Output: ['brand', 'model', 'storage']
```

## 9) items():

**It returns list of tuples representing key-value pairs. [k,v),(k,v),(k,v)]**

```
# Initial dictionary
stock = {"apple": 50, "orange": 100, "grape": 75}
# Get the key-value pairs
all_items = stock.items()
print(all_items)
# Output: dict_items([('apple', 50), ('orange', 100), ('grape', 75)])
```

## 10) copy(): To create exactly duplicate dictionary (cloned copy)

**d1 = d.copy();**

```
# Original dictionary
original = {"a": 1, "b": 2}
# Create a cloned copy
cloned = original.copy()
# Modify the clone
cloned["c"] = 3
print(original) # Output: {'a': 1, 'b': 2} (remains unchanged)
print(cloned)   # Output: {'a': 1, 'b': 2, 'c': 3}
```

**11) setdefault():**

d.setdefault(k,v)

- If the key is already available, then this function returns the corresponding value.
- If the key is not available, then the specified key-value will be added as new item to the dictionary.

```
user_settings = {"theme": "dark", "language": "EN"}
current_theme = user_settings.setdefault("theme", "light")
print(current_theme)   # Output: dark
print(user_settings)   # Output: {'theme': 'dark', 'language': 'EN'}

# Case 2: Key is missing
# It adds the new key-value pair and returns the new value
font_size = user_settings.setdefault("font_size", 12)
print(font_size)       # Output: 12
print(user_settings)   # Output: {'theme': 'dark', 'language': 'EN', 'font_size': 12}
```

## Dictionary Comprehension:

**Comprehension concept applicable for dictionaries also.**

```
squares={x:x*x for x in range(1,6)}
print(squares)
```

Ques:

Write a Program to Enter Name and Percentage Marks in a Dictionary and Display Information on the Screen?

```
d={}
n=int(input("Enter number of students: "))
i=1
while i <=n:
    name=input("Enter Student Name: ")
    marks=input("Enter % of Marks of Student: ")
    d[name]=marks
    i=i+1

for x, y in d.items():
    print(f"Name of Student is {x} and Percentage is {y}")
```

Ques:

**Write a Program to find Number of Occurrences of each Letter present in the given String?**

```
word=input("Enter any word: ")
d={}
for x in word:
    d[x]=d.get(x,0)+1
for k,v in d.items():
    print(k,"occurred ",v," times")
```

**Ques**

**Write a Program to find Number of Occurrences of each Vowel present in the given String?**

```
word=input("Enter any word: ")
vowels={'a','e','i','o','u'}
d={}
for x in word:
    if x in vowels:
        d[x]=d.get(x,0)+1
for k,v in sorted(d.items()):
    print(k,"occurred ",v," times")
```

**Ques**

**Write a Program to take Dictionary from the Keyboard and print the Sum of Values?**

```
d=eval(input("Enter dictionary:"))
s=sum(d.values())
print("Sum= ",s)
```